Introduction to the Laws of Motion

Lesson 3: Newton's First Law

Newton's First Law - Study Notes

Key Concepts

1. **Inertia** – the tendency of an object to resist changes in its state of motion.

2. **State of Rest or Uniform Motion** – an object will stay at rest, or move in a straight line at constant speed, unless acted upon by an external net force.

3. **Net Force** – the vector sum of all forces acting on an object; only a non zero net force can change the object's motion.

4. **Reference Frame** – the law is valid in inertial (non accelerating) frames of reference.

Summary

Newton’s First Law, often called the **law of inertia**, states that a body will maintain its current state—either at rest or moving with constant velocity—unless a net external force acts on it. In everyday life this explains why a parked car does not roll away on its own, and why a passenger feels pushed backward when a vehicle suddenly accelerates. The concept of inertia is a property of mass: the more massive an object, the greater its resistance to changes in motion.

The law also introduces the idea of an **inertial reference frame**—a viewpoint that is itself not accelerating. Within such a frame, the only reason an object’s speed or direction changes is because a net force is applied. Understanding this principle lays the groundwork for the later laws of motion, which quantify how forces produce acceleration.

Important Points

• **Inertia is proportional to mass** – heavier objects have more inertia.

• **No net force !’ no acceleration** (9Gb Ò ’à

• **Balanced forces** (equal magnitude, opposite direction) produce a net force of zero, so the object’s motion does not change.

• **Unbalanced force** (net "` 0) causes acceleration in the direction of the net force.

• The law is **valid only in inertial frames**; accelerating frames require fictitious forces to apply the law.

Examples

| Situation | What the Law Predicts | Explanation |

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| **Parked car on a flat road** | The car remains at rest. | No unbalanced external force is acting horizontally (friction and gravity are balanced). Hence, the net force is zero and the car’s state of rest persists. |

| **Tablecloth trick (quick pull)** | The dishes on the cloth stay (almost) in place while the cloth is pulled away. | The pull is a short, high speed force applied only to the cloth. The dishes have inertia; because the force on them is minimal and the interaction time is brief, their change in motion is negligible, so they stay where they were. |

Quick Review

1. **What is inertia, and how does mass relate to it?**
2. **Why does a passenger feel thrown backward when a car accelerates suddenly?**
3. **If two equal forces act on an object in opposite directions, what is the net force and what does the object do?**
4. **Define an inertial reference frame.**
5. **In the tablecloth trick, why don’t the dishes move with the cloth?**

Further Reading

- Newton’s Second Law (F = ma) – quantifying how net force produces acceleration.
- Newton’s Third Law – action–reaction pairs.
- Friction and its role in real world deviations from ideal inertia.
- Non inertial (accelerating) reference frames and fictitious forces (e.g., centrifugal force).
- Applications of inertia in vehicle safety (seat belts, airbags).